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Saveetha Institute of Medical And Technical Sciences

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Technical Presentation

Comparative Analysis of Polynomial and Logarithmic Algorithms

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Algorithm Overview

Algorithms Considered

Logarithmic Algorithm – O(log n)

- Example: Binary Search
- Divides the search space into halves.

Linear Algorithm – O(n)

- Example: Linear Search
- Checks each element sequentially.

Polynomial Algorithm – $O(n^2)$

- Example: Bubble Sort
- Compares every element with others repeatedly.

Problem Statement

- In today's applications, the amount of data is increasing rapidly, making algorithm efficiency an important factor. Different algorithms require different amounts of time to solve the same problem as the input size grows.
- **Logarithmic (O(log n))**, **Linear (O(n))**, and **Polynomial ($O(n^2)$)** algorithms show different performance characteristics.
- The challenge is to identify which algorithm is more efficient and scalable for different input sizes.
- For this analysis, we assume input sizes ranging from **100 to 100,000 elements.**
- The algorithms are compared based on **execution time, number of operations, memory usage, and scalability.** The goal is to determine the most suitable algorithm for real-world applications by using comparison tables and graphical analysis.

Algorithm / Pseudocode

Binary Search (O(log n))

```
low = 0
high = n-1

while(low <= high)
{
 mid = (low + high)/2

 if(key == a[mid])
     return mid;

 else if(key < a[mid])
     high = mid-1;

 else
     low = mid+1;
}
```

Linear Search (O(n))

```
for(i=0;i<n;i++)
{
   if(a[i]==key)
      return i;
}
```

Bubble Sort (O(n²))

```
for(i=0;i<n-1;i++)
{
   for(j=0;j<n-i-1;j++)
   {
      if(a[j]>a[j+1])
          swap();
   }
}
```

Working Example

Input

```
Array = {5,10,15,20,25,30,35}
Key = 20
```

Binary Search

```
Middle = 20

Found in 1 comparison
```

Linear Search

```
5
10
15
20

Found after 4 comparisons
```

Bubble Sort

Input	Output
5 3 2 4	2 3 4 5

Complexity Analysis

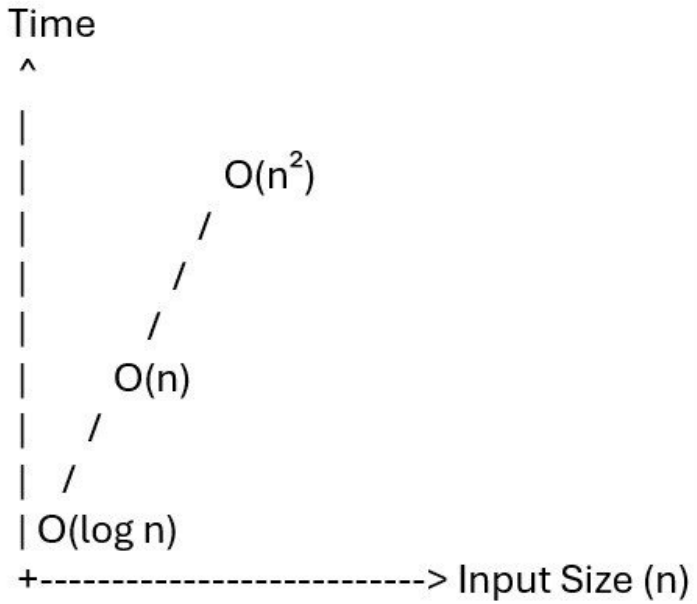
Algorithm	Best	Average	Worst	Space
Binary Search	O(1)	O(log n)	O(log n)	O(1)
Linear Search	O(1)	O(n)	O(n)	O(1)
Bubble Sort	O(n)	$O(n^2)$	$O(n^2)$	O(1)

Comparative Analysis

Feature	O(log n)	O(n)	$O(n^2)$
Example	Binary Search	Linear Search	Bubble Sort
Scalability	Excellent	Good	Poor
Performance	Fastest	Moderate	Slow
Large Input	Best	Acceptable	Not Recommended
Time Growth	Very Slow	Linear	Rapid

Performance Graph

- The **X-axis** represents the **Input Size (n)**, and the **Y-axis** represents the **Execution Time** of the algorithms.
- The **O(log n)** curve grows very slowly, showing that logarithmic algorithms are the **most efficient** and perform well even for large datasets.
- The **O(n)** curve increases at a constant rate with the input size, providing **moderate performance** and good scalability. The **$O(n^2)$** curve rises steeply as the input size increases, indicating that polynomial algorithms become **less efficient** for large datasets.
- **Overall, the graph shows that O(log n) is the best choice for large inputs, O(n) offers acceptable performance, and $O(n^2)$ is suitable only for small datasets due to its higher execution time.**



Applications

Binary Search

- Database searching
- Dictionary search
- Phone contacts
- Search engines

Linear Search

- Small datasets
- Unsorted lists
- File searching

Bubble Sort

- Educational purposes
- Small datasets
- Algorithm learning

Applications

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Results & Discussion

Result

- **Binary Search (O(log n))** provides the best performance but requires sorted data.
- **Linear Search (O(n))** is simple and works well for unsorted or small datasets.
- **Bubble Sort ($O(n^2)$)** becomes inefficient as the input size grows.

Discussion

- For large datasets, logarithmic algorithms are highly scalable.
- Linear algorithms offer a balance between simplicity and performance.
- Polynomial algorithms should generally be avoided for large-scale problems

Conclusion & References

Conclusion

- Time complexity significantly impacts algorithm performance.
- **O(log n)** algorithms are the most scalable.
- **O(n)** algorithms are suitable for moderate-sized problems.
- **$O(n^2)$** algorithms are mainly useful for learning and small datasets.

References (IEEE)

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- M. T. Goodrich, R. Tamassia, and M. H. Goldwasser, *Data Structures and Algorithms in Java*, 7th ed., Wiley, 2022.
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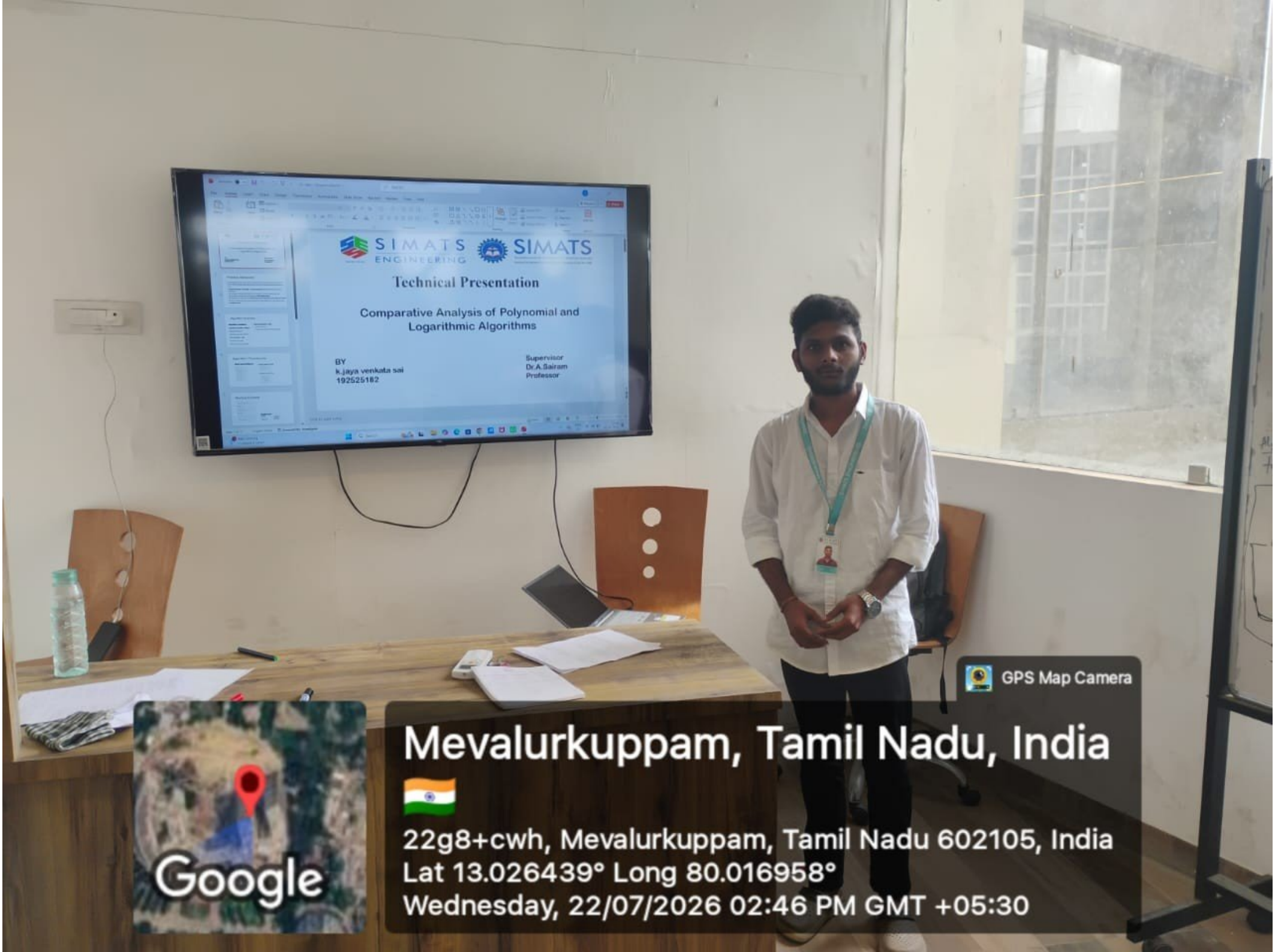
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